Mark schemes

Q1.

(a) (i) $y = (4x^2 + 3x + 2)^{10}$

$$\frac{dy}{dx} = 10(4x^2 + 3x + 2)^9 (8x + 3)$$

M1

For f(x)( )^9 where f(x) ≠ k and is linear

A1

2

(ii) $y = x^2 \tan x$

Product rule

M1

$$\frac{dy}{dx} = x^2 \sec^2 x + 2x \tan x$$

A1

2

(b) (i) $x = 2y^3 + \ln y$

$$\frac{dx}{dy} = 6y^2 + \frac{1}{y}$$

B1

1

(ii) At (2,1)

$$\frac{dx}{dy} = 6 + 1 = 7$$

M1

$$\frac{dy}{dx} = \frac{1}{7}$$

May be implied

A1 √

$$(y - 1) = \frac{1}{7}(x - 2)$$

OE

A1 3

[8]

Q2.

(a) (i) $\int(4-e^{2x})dx$

$= 4x - \frac{1}{2}e^{2x} (+c)$

4x

B1

$-\frac{1}{2}e^{2x}$

B1 2

(ii) $\int_0^{\ln 2} = \left[4x - \frac{1}{2}e^{2x}\right]_0^{\ln 2}$

$= \left[4\ln 2 - \frac{1}{2}e^{2\ln 2}\right] - \left[(0) - \frac{1}{2}(e^{0})\right]$

Substitute both ln 2 and 0 correctly into an integrated expression

M1

$= 4\ln 2 - 2 + \frac{1}{2}$

Convincing

$= 4\ln 2 - \frac{3}{2}$

AG

A1 2

(b) (i) $x = 0$

$y = 4 - 1 = \underline{\underline{3}}$

B1 1

(ii) At B, $y = 0$

$4 - e^{2x} = 0$

Or reverse argument

M1

$e^{2x} = 4$

$x = \ln 2$

AG

A1

2

(c) $\frac{dy}{dx} = -2e^{2x}$

B1

$x = \ln 2$, Gradient $= -2e^{2\ln 2} = -8$

x = ln 2 into ke^{2x}

M1

Gradient normal = $\frac{1}{8} = \frac{1}{2e^{2\ln 2}}$

OE

A1

Equation $y = -\frac{1}{8}x - \frac{1}{8}\ln 2$

OE

A1

4

(d) When $x = 0$

$y = -\frac{1}{8}\ln 2$

Attempt to integrate their line and substitute
x = 0, ln 2

M1

Area Δ = $\frac{1}{16}(\ln 2)^2$ condone – ve sign = 0.03

$\frac{1}{2}$ (their y) × ln 2

A1 ✓

Total area $4\ln 2 - \frac{3}{2} + \frac{1}{16}(\ln 2)^2 = 1.30$ AWRT

CSO

A1

3

[14]

Q3.

(a) (i) $\int \frac{dy}{y} = \int \sin t \, dt$

Attempt to separate and integrate

M1

$\ln y = -\cos t + c$

A1 for ln y; A1 for $-\cos t$; condone missing c

A1,A1

$y = ae^{-\cos t}$

A present; or $y = e^{-\cos t + C}$

A1

4

(ii) $y = 50$, $t = \pi$: $50 = Ae^{-\cos \pi} = Ae$

Substitute $y = 50$, $t = \pi$ to find constant
Can have $50 = e^{1+C}$ if substituted in above

$e^C = \frac{50}{e}$

M1
A1

$y = 50e^{-1}e^{-\cos t}$

AG (convincingly obtained)

A1

3

Alternative:
Must have a constant in answer to (a)(i)

$y = Ae^{-\cos t}$ or $y = e^{-\cos t + c}$ or $\ln y = -\cos t + c$

$50 = Ae^{-\cos\pi}$ $50 = e^{-\cos\pi + c}$ $\ln 50 = -\cos\pi + c$

(M1)

$50 = Ae$ $50 = e^{1+c}$ $\ln y = -\cos t + \ln 50 - 1$

(A1)

$y = 50e^{-1-\cos t}$ $y = e^{-\cos t}\dfrac{50}{e}$ $\ln\left(\dfrac{y}{50}\right) = -1 - \cos t$

(A1)

Alternative:

Substitute $y = 50$, $t = \pi$ into $\ln y = -\cos t + c$ *M1*

$\ln y = -\cos t + \ln 50 - 1$ *A1*

$\ln\dfrac{y}{50} = -1 - \cos t$ *(AG)* *A1*

(b) (i) $t = 6$: $y = 50e^{-1}e^{-\cos 6} = 7.0417\ldots \approx 7\text{cm}$

Degrees 6.8 SC1
7 or 7.0 for A1

M1A1

2

(ii) $t = \pi \Rightarrow (\sin t = 0 \Rightarrow)\dfrac{dy}{dt} = 0$

Condone x for t

B1

$\dfrac{d^2y}{dt^2} = y\cos t + \dfrac{dy}{dt}\sin t$

For attempt at product rule including $\dfrac{dy}{dt}$ *term; must have* $\dfrac{d^2y}{dt^2} =$

M1

$t = \pi$

A1

$$\frac{d^2y}{dt^2} = y\cos\pi + \frac{dy}{dt}\sin\pi$$

$= -50 \Rightarrow$ max

Accept = – y, with explanation that y is never negative

A1

4

Alternative:

$$y = 50e^{-(1+\cos t)} = \frac{50}{e}e^{-\cos t}$$

$$\frac{dy}{dt} = \frac{50}{e}e^{-\cos t} \times \sin t = 0 \text{ at } t = \pi$$

(B1)

$$\frac{d^2y}{dt^2} = \frac{50}{e}e^{-\cos t} \times \cos t + \frac{50}{e}e^{-\cos t} \times \sin^2 t$$

Attempt at product rule

(M1)

Correct

(A1)

Substitute $t = \pi \rightarrow -50 \Rightarrow$ max

(A1)

[13]

Q4.

(a) (i) $t^3 - 52t + 96$

one term correct

M1

another term correct

A1

all correct (no + c etc)

A1

3

(ii) $3t^2 - 52$

ft one term correct

M1

ft all "correct"

A1 √

2

(b) $\frac{dy}{dt} = 8 - 104 + 96$

substitute t = 2 into their $\frac{dy}{dt}$

M1

= 0 $\Rightarrow$ stationary value

CSO; shown = 0 + statement

A1

Substitute t = 2 into $\frac{d^2y}{dt^2}$ (= – 40)

any appropriate test, e.g. $y'(1)$ and $y'(3)$

M1

$\frac{d^2y}{dt^2} < 0 \Rightarrow$ max value

all values (if stated) must be correct

A1

4

(c) Substitute t = 1 into their $\frac{dy}{dt}$

must be their $\frac{dy}{dt}$ *NOT* $\frac{d^2y}{dt^2}$

M1

Rate of change = 45 (cm s^{-1})

ft their $y'(1)$

A1 √

2

(d) Substitute t = 3 into their $\frac{dy}{dt}$

interpreting their value of $\frac{dy}{dt}$

M1

(27 – 156 + 96 = 33 < 0)

$\Rightarrow$ decreasing when t = 3

allow increasing if their $\frac{dy}{dt}$ *> 0*

E1 √

2

[13]

Q5.

(a) $y_P = 4$

B1

1

(b) $y = 1 + \frac{2}{x} + \frac{2}{x} + \frac{4}{x^2}$

(B1 if only one error in the expansion)
For B2 the last line of the candidate's solution must be correct

B2,1,0

$y = 1 + 4x^{-1} + 4x^{-2}$

2

(c) $\frac{dy}{dx} = -4x^2 - 8x^{-3}$

Index reduced by 1 after differentiating x to a negative power

M1

At least 1 term in x correct ft on expn

A1ft

CSO Full correct solution. ACF

A1

3

(d) When x = 2, $\frac{dy}{dx} = -4 \times 2^{-2} - 8 \times 2^{-3}$

Attempt to find $y'(2)$.

M1

Gradient = – 1 – 1 = – 2

AG (be convinced-no errors seen)

A1

2

(e) $-2 \times m' = -1$

$m_1 \times m_2 = -1$ OE stated or used. PI

M1

$y - 4 = m(x - 2)$

C's y_P from part (a) if not recovered;
m must be numerical.

M1

$$y - 4 = \frac{1}{2}(x - 2)$$

Ft on candidate's y_P from part (a) if not recovered.

A1ft

$x - 2y + 6 = 0$

CAO Must be this or $0 = x - 2y + 6$

A1

4

[12]

Q6.

(a) $y = \ln x$

penalise + c once on 1 (a) or 2 (a)

$$\frac{dy}{dx} = \frac{1}{x}$$

B1

1

(b) $y = (x + 1) \ln x$

$$\frac{dy}{dx} = (x+1) \times \frac{1}{x} + \ln x$$

product rule

M1
A1
2

(c) $y = (x + 1)\ln x$

$\frac{dy}{dx} = \frac{1}{x} + 1 + \ln x$

$x = 1$: $\frac{dy}{dx} = 1 + 1 = 2$

substitute $x = 1$ into their $\frac{dy}{dx}$

M1

Grad normal $= -\frac{1}{2}$

use of $m_1 m_2 = -1$

M1

CSO

A1

$y = -\frac{1}{2}(x - 1)$

OE

A1
4
[7]

Q7.

(a) (i) $t = 0$: $x = 3$

B1
1

(ii) $t = 14$: $x = 15 - 12e^{-1}$

or $15 - 12e^{\frac{-14}{14}}$

M1

= 10.6

CAO

A1

2

(b) (i) $-5 = -12e^{-\frac{t}{14}}$

substitute x = 10; rearrange to form p = $qe^{-\frac{t}{14}}$

M1

$\ln\left(\frac{5}{12}\right) = -\frac{t}{14}$ (OE)

take lns correctly

m1

$t = 14\ln\left(\frac{12}{5}\right)$

must come from correct working

A1

3

(ii) t = 12.256... ≈ 12 days

ft on a, b if a > b; accept t = 12 NMS
Accept 12 from incorrect working in (b)(i)
Accept 13 if 12.2 or 12.3 seen

B1F

1

(c) (i) $\frac{dx}{dt} = -\frac{1}{14}x - 12e^{-\frac{t}{14}}$

differentiate; allow sign error

condone $\frac{dy}{dx}$ used consistently

M1

$= -\frac{1}{14}(x - 15)$

Or $\frac{1}{14}\left(12e^{-\frac{t}{14}}\right)$ *and* $12e^{-\frac{t}{14}}$ *= 15 − x seen*

m1

$= \frac{1}{14}(15 - x)$

AG − be convinced CSO

A1

3

Alt: $t = -14\ln\left(\frac{15-x}{12}\right)$

attempt to solve given equation for t

(M1)

$$\frac{dt}{dx} = \frac{-14\left(-\frac{1}{12}\right)}{\left(\frac{15-x}{12}\right)}$$

differentiate wrt x, with $\frac{1}{\frac{15-x}{12}}$ *seen; OE*

(m1)

$$\frac{dt}{dx} = \frac{14}{15-x} \Rightarrow \frac{dx}{dt} = \frac{1}{14}(15-x)$$

AG − be convinced

(A1)

(3)

Alt: (backwards)

$$\int \frac{dx}{15-x} = \int \frac{dt}{14} = \pm 14\ln(15-x) = t + c$$

(M1)

Use (0,3): $-14 \ln (15 - x) + 14 \ln 12 = t$

(m1)

Solve for x: $x = 15 - 12e^{-\frac{t}{14}}$

All steps shown

(A1)

(3)

(ii) rate of growth = 0.5 (cm per day)

Accept $\frac{7}{14}$

B1

1

[11]

Q8.

(a) $x = 1$, $5a^2 - a - 4 = 0$

condone y for a

M1

$(5a+4)(a-1)=0,\ a=1$

AG – be convinced, both factors seen

or $a = -\frac{4}{5}$ *or 1* $\Rightarrow$ *a = 1*

A0 for 2 positive roots

(substitute (1, 1) $\Rightarrow$ *5 = 5 no marks)*

A1

2

(b) $\frac{dy}{dx} + 4$

(Ignore ' $\frac{dy}{dx}$ *=' if not used, otherwise loses final A1)*

B1B1

$= 10xy^2 + 10x^2y\frac{dy}{dx}$

attempt product rule, see two terms added

M1

chain rule, $\frac{dy}{dx}$ *attached to one term only*

M1

condone 5 × 2 for 10

A1

$x = 1, y = 1$ $\quad \frac{dy}{dx} + 4 = 10 + 10\frac{dy}{dx}$

two terms, or more, in $\frac{dy}{dx}$

M1

$\frac{dy}{dx} = -\frac{6}{9} = \left(-\frac{2}{3}\right)$

CSO

A1

7

Alt (for last two marks)

$\frac{dy}{dx} = \frac{10xy^2 - 4}{1 - 10x^2y}$

find $\frac{dy}{dx}$ in terms of x, y and substitute x = 1, y = 1 must be from expression with two terms or more in $\frac{dy}{dx}$

(M1)

(1, 1) $\Rightarrow \frac{10 - 4}{1 - 10} = -\frac{6}{9}$

(A1)

(c) $\frac{y - 1}{x - 1} = -\frac{2}{3}$ (OE)

ft on gradient
ISW after any correct form

B1F

1

[10]

Q9.

(a) (i) $\frac{dV}{dt} = 2t^5 - 8t^3 + 6t$

One term correct unsimplified

M1

Further term correct unsimplified

A1

All correct unsimplified (no + c etc)

A1

3

(ii) $\frac{d^2V}{dt^2}$ $= 10t^4 - 24t^2 + 6$

One term FT correct unsimplified

M1

***CSO**. All correct simplified*

A1

2

(b) Substitute $t = 2$ into their $\frac{dV}{dt}$

M1

(= 64 − 64 + 12) = 12

***CSO**. Rate of change of volume is 12 m^3 s^{-1}*

A1

2

(c) (i) $t = 1 \Rightarrow \frac{dV}{dt} = 2 - 8 + 6$

Or putting their $\frac{dV}{dt}$ *= 0*

M1

$= 0 \Rightarrow$ Stationary value

***CSO**. Shown to = 0 **AND** statement*

(If solving equation must obtain t = 1)

A1

2

(ii) $t = 1 \Rightarrow \frac{d^2V}{dt^2} = -8$

Sub t = 1 into their second derivative or equivalent full test.

M1

Maximum value

Ft if their test implies minimum

A1 √

2

[11]

Q10.

(a) $y_D = 3 + 1 = 4$ or $y_C = 12 - 8 = 4$

Attempt at either y coordinate M1

Area $ABCD = 3 \times 4 = 12$ A1 **2**

(b) (i) $x^3 - \frac{x^4}{4}$ $(+C)$

Increase one power by 1 M1

One term correct unsimplified A1

All correct unsimplified (condone no +C) A1 **3**

(ii) Sub limits −1 and 2 into their (b)(i) ans

May use both −1, 0 and 0, 2 instead M1

$[8 - 4] - \left[-1 - \frac{1}{4}\right]$ $= 5\frac{1}{4}$ A1

Shaded area = "their" (rectangle− integral)

Alt method: difference of two integrals M1

$= 12 - 5\frac{1}{4} = 6\frac{3}{4}$

CSO*. Attempted M2, A2* A1 **4**

(c) (i) $\frac{dy}{dx} = 6x - 3x^2$

One term correct M1

All correct (no +C etc)

A1

2

(ii) When x = 1, y = 2 when x = 1,

May be implied by correct tgt equation

B1

$\frac{dy}{dx}$ = 3 as 'their' grad of tgt

Ft their derivative when x = 1

M1ft

Tangent is $y - 2 = 3(x - 1)$

Any correct form $y = 3x - 1$ etc

A1

3

(iii) Decreasing when $\frac{dy}{dx} = 6x - 3x^2 < 0$

Watch no fudging here!!
May work backwards in proof.

M1

$3(2x - x^2) < 0 \quad \Rightarrow x^2 - 2x > 0$

***AG** (be convinced no step incorrect)*

A1

2

(d) Two critical points 0 and 2

Marked on diagram or in solution

M1

$x > 2$, $x < 0$ ONLY

or M1 A0 for $0 < x < 2$ or $0 > x > 2$

***SC** B1 for $x > 2$ (or $x < 0$)*

A1

2

[18]

Q11.

$y'(x) = 16 - x^{-2}$

One term correct

M1

Both correct

A1

$y'(x) = 16 - \frac{1}{x^2}$

$x^{-2} = \frac{1}{x^2}$ *OE PI*

B1

$y'(x) = 0 \Rightarrow 16x^2 = 1$;

c's $y'(x) = 0$ and one relevant further step

M1

$\Rightarrow x = \pm\frac{1}{4}$

Both answers required.

A1 5

[5]

Q12.

(a) $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} - 3$

One term correct

M1

Both correct

A1 2

(b) (i) When $x = 0$, $\frac{dy}{dx} = -3$

Ft provided answer < 0.

B1F √

Eqn of tangent at O is $y = -3x$

OE Ft on $y'(0)$

B1F √ 2

(ii) At (9,0) $\frac{dy}{dx} = \frac{3}{2}(9)^{\frac{1}{2}} - 3$

Attempt to find y'(9)

M1

Eqn tangent at A is $y - 0 = y'(9)[x - 9]$

OE

m1

$\Rightarrow y = \frac{3}{2}(x - 9) \Rightarrow 2y = 3x - 27$

***CSO**. AG*

A1

3

(iii) Eliminating $y \Rightarrow -6x = 3x - 27$

OE method to one variable
(eg $2y = -y - 27$)

M1

$9x = 27 \Rightarrow x = 3$

[A1F for each coordinate; only ft on $y = kx$ tangent in (b)(i) for $k < 0$]

A1F

When $x = 3$, $y = -9${. $P(3, -9)$}

A1F

3

(c) $\int\left(x^{\frac{3}{2}} - 3x\right) dx = \frac{2}{5}x^{\frac{5}{2}} - \frac{3x^2}{2}$ $(+c)$

One power correct

M1

Condone absence of "+ c" and unsimplified forms

A2,1,0

3

(d) $\int_0^9\left(x^{\frac{3}{2}} - 3x\right) dx =$

PI

B1

$= \frac{2}{5} \times 9^{\frac{5}{2}} - \frac{3}{2} \times 9^2 - 0$

Correct use of limits following integration

M1

= − 24.3

Area of triangle OPA = $\frac{1}{2}$ × 9 × $|y_P|$

M1

Sh.Area = $\frac{1}{2}$ × 9 × $|y_P|$ − $\left|\int_0^9 \left(x^{\frac{3}{2}} - 3x\right) dx\right|$

OE

M1

= 40.5 − 24.3 = 16.2

A1 5

[18]

Q13.

(a) $\frac{dy}{dx}$ = 3sec² $3x$

M1

for sec 3x

SC / 3sec² x B1

A1

Alternative

Use of product / Quotient rule (M1)

Good attempt

$\frac{3\cos^2 3x + 3\sin^2 3x}{\cos^2 3x}$ (A1)

Correct

2

(b) $\frac{dy}{dx} = \frac{(2x+1)3 - 2(3x+1)}{(2x+1)^2} = \frac{6x+3-6x-2}{(2x+1)^2}$

use of quotient rule

M1 A1

$= \frac{1}{(2x+1)^2}$

AG (no errors)

A1

Alternative

$-2(3x + 1)(2x + 1)^{-2} + 3(2x + 1)^{-1}$ (M1A1)

$= \frac{1}{(2x+1)^2}$ (A1)

$y = \frac{3}{2} - \frac{1}{2}(2x + 1)^{-1}$ *M1A1*

$\frac{dy}{dx} = (2x + 1)^{-2}$ *A1*

$= \frac{1}{(2x+1)^2}$ *AG*

3

[5]

Q14.

(a) (i) $f' = \frac{dy}{dx} = 4x^3 + 2$

B1

1

(ii) $\int \frac{2x^3 + 1}{x^4 + 2x} dx$

$= \frac{1}{2}\ln(x^4 + 2x)(+c)$

For k ln $(x^4 + 2x)$

M1

By substitution k ln u *M1*

correct *A1*

A1

2

(b) (i) $u = 2x + 1$

$du = 2\,dx$

B1

$\int x\sqrt{2x+1}\,dx =$

$\int \left(\frac{u-1}{2}\right)\sqrt{u}\ \frac{du}{2}$

Must be in terms of u only incl. du

M1

$= \frac{1}{4}\int\left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right)du$

AG

A1

3

(ii) $\int_0^4 dx = \int_1^9 du$

Or changing u's to x's at end

B1

$\frac{1}{4}\int u^{\frac{3}{2}} - u^{\frac{1}{2}} = \frac{1}{4}\left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} - \frac{u^{\frac{3}{2}}}{\frac{3}{2}}\right]$

M1
A1

$= \frac{1}{4}\left[\left(\frac{2}{5}(9)^{\frac{5}{2}} - \frac{2}{3}(9)^{\frac{3}{2}}\right) - \left(\frac{2}{5} - \frac{2}{3}\right)\right]$

$= \frac{1}{4}\ [79.2 + 0.2\dot{6}\ ]$

$= 19.86$

Sight of any of these 3 lines

$= 19.9$

AG

A1

4

[10]

Q15.

(a) $\frac{dy}{dx} = -10x^4$

kx^4 condone extra term

M1

Correct derivative unsimplified

A1

2

(b) When $x = 1$, gradient = −10

FT their gradient when $x = 1$

B1 √

Tangent is

Attempt at y & tangent (not normal)

M1

$y - 5 = -10(x - 1)$ *or* $y + 10x = 15$ etc

CSO Any correct form

A1

3

(c) When $x = -2$ $\frac{dy}{dx} = -160$ (or < 0)

Value of their $\frac{dy}{dx}$ when $x = -2$

B1 √

($\frac{dy}{dx}$ < 0 hence) y is **decreasing**

ft Increasing if their $\frac{dy}{dx} > 0$

E1 √

2

[7]

Q16.

(a) (i) $\frac{dy}{dx} = 3x^2 - 20x + 28$

One term correct

M1

Another term correct

A1

All correct (no + c etc)

A1

3

(ii) Their $\frac{dy}{dx}$ = 0 for stationary point

Or realising condition for stationary point

M1

$(x - 2)(3x - 14) = 0$

Attempt to solve using formula/ factorise

m1

$\Rightarrow x = 2$

A1

or $x = \frac{14}{3}$

Award M1, A1 for verification that $x = 2 \Rightarrow \frac{dy}{dx} = 0$ then may earn m1 later

A1

4

(b) (i) $\frac{x^4}{4} - \frac{10x^3}{3} + 14x^2 \quad (+ c)$

One term correct unsimplified

M1

Another term correct unsimplified

A1

All correct unsimplified (condone missing + c)

A1

3

(ii) $\left[\frac{81}{4} - 90 + 126\right] \quad (-0)$

Attempt to sub limit 3 into their (b)(i)

M1

$= 56\frac{1}{4}$

AG Integration, limit subtraction all correct

A1

2

(iii) Area of triangle = $31\frac{1}{2}$

Correct unsimplified $\frac{1}{2} \times 21 \times 3$

B1

Shaded Area = $56\frac{1}{4}$ − triangle area

M1

$= 24\frac{3}{4}$

Or equivalent such as $\frac{99}{4}$

B1

3

[15]

Q17.

(a) (i) When $x = 4$, $\frac{dy}{dx} = 3(2) + \frac{16}{16} - 7 = 0$

AG Be convinced

B1

1

(ii) $\frac{16}{x^2} = 16x^{-2}$

Accept $k = -2$

B1

1

(iii) $\frac{d^2y}{dx^2} = 3 \times \frac{1}{2}x^{-\frac{1}{2}}$ + 16 × (−2)x^{-3} − 0

A power decreased by 1

M1

$\frac{d^2y}{dx^2} = \frac{3}{2}x^{-\frac{1}{2}}$; $-32x^{-3}$

candidate's negative integer k [−1 for >2 term(s)]

A1;
A1 √

3

(iv) When x = 4, $\frac{d^2y}{dx^2} = \frac{3}{4} - \frac{32}{64} = \frac{1}{4}$

Attempt to find y''(4) reaching as far as two simplified terms

M1

Minimum since $y''(4) > 0$

candidate's sign of y''(4)

E1 √

2

[**Alternative:** Finds the sign of $y'(x)$ either side of the point where x = 4, need evidence rather than just a statement: (M1)
Correct ft conclusion with valid reason E1 √]
[In both, condone absent statement $y'(4) = 0$]

(b) (i) At $P(1,8)$, $\frac{dy}{dx}$ = $3(1)^{\frac{1}{2}} + \frac{16}{1^2}$ − 7 = 12

AG Be convinced

B1

1

(ii) Gradient of normal = $-\frac{1}{12}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 8 = m[x - 1]$

Can be awarded even if m = 12

M1

$y - 8 = -\frac{1}{12}(x - 1) \Rightarrow 12y - 96 = -x + 1 \Rightarrow 12y + x = 97$

Any correct form of the equation

A1

3

(c) (i) $\int 3x^{\frac{1}{2}} + \frac{16}{x^2} - 7\,dx = \dots\dots\dots = 3\frac{x^{\frac{3}{2}}}{\frac{3}{2}} + 16\frac{x^{-1}}{-1} - 7x + c$

One power correct.

M1

A1 if 2 of 3 terms correct candidate's negative integer k

Condone absence of "+ c"

A2,1,0 √

3

(ii) $y = 2x^{\frac{3}{2}} - 16x^{-1} - 7x + c$ (*)

y = candidate's answer to (c)(i) with tidied coefficients and with '+c'. ('y =' PI by next line)

B1 √

When $x = 1$, $y = 8 \Rightarrow 8 = 2 - 16 - 7 + c$

Substitute. (1,8) in attempt to find constant of integration

M1

$y = 2x^{\frac{3}{2}} - 16x^{-1} - 7x + 29$

Accept c = 29 after (*), including y =, stated

A1

3

[17]

Q18.

(a) $y = e^{2x} - 10e^{x} + 12x$

(i) $\frac{dy}{dx} = 2e^{2x} - 10e^{x} + 12$

$2e^{2x}$

B1

remaining terms correct, no extras

B1

2

(ii) $\frac{d^2y}{dx^2}$ = $4e^{2x} - 10e^{x}$

ft 1 slip

B1F

1

(b) (i) $2e^{2x} - 10e^{x} + 12 = 0$

$e^{2x} - 5e^{x} + 6 = 0$

AG (be convinced)

B1

1

(ii) $z^2 - 5z + 6 = 0$

use of z = e^{x} oe

M1

z = 2, 3

z = 2, e^{x} = 2

x = ln 2

finding e^{x} = their 2,3

M1

z = 3, $e^{x} = 3$

x = ln 3

all correct AG

A1

SC: verification

$\ln 2$ *(B1)*

$\ln 3$ *(B1)*

3

(iii) x = ln 2:

$y = e^{2\ln 2} - 10e^{\ln 2} + 12\ln 2$

either substitution of their x = ln 2 $\left(e^x = 2\right)$

or their x = ln 3 $\left(e^x = 3\right)$

M1

or
$2^2 - 10 \times 2 + 12 \ln 2 = 4 - 20 + 12 \ln 2 = -16 + 12 \ln 2$

A1

$x = \ln 3$:

$y = e^{2\ln 3} - 10e^{\ln 3} + 12\ln 3$ = 9 − 30 + 12 ln 3 = − 21 + 12 ln 3

A1

3

(iv) $x = \ln 2$:

$\frac{d^2y}{dx^2} = 4e^{2\ln 2} - 10e^{\ln 2}$

use of; in either of their $e^x = 2, 3$ *into their* $\frac{d^2y}{dx^2}$

M1

= 16 − 20 = −4
$\therefore$ maximum
CSO

A1

$x = \ln 3$:

$\frac{d^2y}{dx^2} = 4e^{2\ln 3} - 10e^{\ln 3}$

= 36 − 30 = 6
$\therefore$ minimum
CSO

A1

3

[13]

Q19.

(a) $y = x \ln x$

$\frac{dy}{dx} = x \times \frac{1}{x} + \ln x$

use of product rule (only differentiating, 2 terms with + sign)

M1

$= \ln x + 1$

A1

2

(b) $\int (\ln x + 1)dx = x \ln x$

OE; attempt at parts with

M1

$\int \ln x \, dx = x \ln x - x (+c)$

A1

2

(c) $\int_1^5 \ln x \, dx = [x \ln x - x]_1^5$

$= (5 \ln 5 - 5) - (1 \ln 1 - 1)$

correct substitution of limits into their (b) provided $\ln x$ *is involved*

M1

$5 \ln 5 - 4$

ISW

A1

2

[6]

Q20.

(a) $x = \frac{1}{2}$ $\quad y = \frac{\pi}{2}$ $\left(\text{or } 1.57, \sin^{-1} 1\right)$

ignore 90°

B1

1

(b) (i) $y = \sin^{-1} 2x$

$\sin y = 2x$ and $\frac{1}{2}\sin y = x$

AG (be convinced)

B1 **1**

(ii) $\frac{dx}{dy} = \frac{1}{2}\cos y$

B1 **1**

(c) $\frac{dy}{dx} = \frac{2}{\cos y}$

M1 for $\frac{k}{\cos y}$

M1A1

$\sin y = 2x$ and $\sin^2 + \cos^2 = 1$

use of to get $\cos y$ or $\cos^2 y$

M1

$\cos y = \sqrt{1-4x^2}$

$\frac{dy}{dx} = \frac{2}{\sqrt{1-4x^2}}$

AG; condone omission of proof of sign

A1 **4**

[7]

Q21.

(a) $x = 1$ $y^2 - y + 3 - 5 = 0$

M1

$(y-2)(y+1) = 0$

Attempt to solve quadratic equation with x = 1

M1

$y = 2 \quad y = -1$

A1

3

(b) (i) $2y\frac{dy}{dx} - x\frac{dy}{dx} - y + 6x = 0$

+6x; −5 $\to 0$

B1B1

Chain rule

B1

Product rule (M1 two terms)

M1A1

$6x - y + (2y - x)\frac{dy}{dx} = 0$

Factorise and obtain answer given

A1

6

Alternative

$\frac{dy}{dx}(y - x)^2$ = $(y - x)(0 - 6x)$

$5 \to 0$

(B1)

−6x

(B1)

$-(5 - 3x^2)\left(\frac{dy}{dx} - 1\right)$

Recognisable attempt at quotient rule

(M1)

Completely correct OE

(A1)

$\frac{dy}{dx}\left[(y + x)^2 + (5 - 3x^2)\right] = (y - x)(-6x)$

$+ (5 - 3x^2)$

Factorise out $\frac{dy}{dx}$

(A1)

Given answer

Correct answer from correct working

Be convinced

(A1)

(ii) $(1, 2)$ $\quad \frac{dy}{dx} = -\frac{4}{3}$

Substitute x = 1 and one y value from (a)

M1

$(1, -1)$ $\quad \frac{dy}{dx} = \frac{7}{3}$

Both; follow on candidates y s

OE $\frac{-7}{-3}$ *; 3SF*

A1F

2

(iii) $y - 6x = 0$

B1

$(6x)^2 - x \times 6x + 3x^2 - 5 = 0$

M1

$36x^2 - 6x^2 + 3x^2 - 5 = 0$

$33x^2 - 5 = 0$

AG convincingly obtained

A1

3

[14]

Q22.

(a) $\frac{dy}{dx} = 5x^4 - 18x^2 - 3$

Decrease one power by 1

M1

One term correct

A1

All correct

A1

3

(b) (i) Sub $x = 2$ into their $\frac{dy}{dx}$

80 – 72 – 3

M1

Shown to equal 5

AG (be convinced)

A1

2

(ii) Gradient of normal $= -\frac{1}{5}\left(y + \frac{1}{5}x + \ldots\right)$

Or $m_1 m_2 = -1$ used or stated

B1

$y - 3 = -\frac{1}{5}(x - 2)$

Trying normal NOT tangent
or y = mx + c and attempt to find c

M1

$x + 5y = 17$ (integer coefficients)

Or integer multiple of coefficients

A1

3

(c) Sub $x = 1$ into their $\frac{dy}{dx}$ (= –16 < 0)

(5 – 18 –3 = –16) (Watch $\frac{d^2y}{dx^2} = -16$ *!)*

M1

Negative value $\Rightarrow$ DECREASING

Correct interpretation of sign of $\frac{dy}{dx}$

E1ft

2

[10]

Q23.

(a) Sides $24 - 2x$, $9 - 2x$

Either correct

B1

$V = x(24 - 2x)(9 - 2x)$

3 sides involving x multiplied together

M1

$= 4x^3 - 66x^2 + 216x$

AG (be convinced)

A1

3

(b) (i) $\frac{dV}{dx} = 12x^2 - 132x + 216$

Power decreased by 1

M1

One term correct

A1

All correct (no +C etc)

A1

3

(ii) Putting their $\frac{dV}{dx} = 0$ (must see this first)

Or their $12x^2 - 132x + 216 = 0$

Or $12(x^2 - 11x + 18) = 0$ or statement

M1

$\Rightarrow x^2 - 11x + 18 = 0$

AG (be convinced)

A1

2

(iii) $(x - 2)(x - 9) = 0$

M1

Factors, comp sq or formulae used (1 slip)
$x = 2, x = 9$

A1

2

(iv) Reject $x = 9$, since $9 - 2x < 0$
$x = 2$ *is only possible value*

E1

1

(c) (i) $\frac{d^2V}{dx^2} = 24x - 132$

Differentiating their $\frac{dV}{dx}$ *(eg* $2x - 11$*)*

M1

Correct

A1

2

(ii) $x = 2 \text{ only} \Rightarrow \frac{d^2V}{dx^2} = -84$ (or < 0)

Correct $\frac{d^2V}{dx^2}$ *value OE full test.*

B1

Maximum value
ft if their test implies minimum

E1✓

2

[15]

Q24.

(a) (i) $y = x + 2x^{-1}$

PI by sight of $-2x^{-2}$

B1

$\frac{dy}{dx} = 1 - 2x^{-2}$

One term correct

M1

OE

A1

3

(ii) When $x = 2$, $\frac{dy}{dx} = 1 - \frac{2}{4} = \frac{1}{2}$

CSO AG (be convinced)

A1

1

(b) When $x = 2$, $y = 3$

For y = 3

B1

gradient of normal = − 2

$m \times m'$ *= –1 used*

M1

Equation normal $y - 3 = -2(x - 2)$

$y - "3" = m(x - 2)$ *OE*

M1

Award at 1st correct form

A1

4

[8]

Q25.

(a) (i) $(2 + x)^3 =$

M1

$(2^3) + 3(2^2)(x) + 3(2)(x^2) + (x^3)$

A1

... = 8 + 12x + 6x2 + x3 (*)

Any valid method; must contain all components

Accept a = 12
Accept b = 6

A1 3

(ii) $(2-x)^3 = 8 - 12x + 6x^2 - x^3$ (**)
Clear $x \to -x$ in (i) OE

M1

ft numerical a and b

A1ft 2

(b) $(2+x)^3 - (2-x)^3$ = (*) – (**)
Subtracts the 2 expressions in (a)

M1

…… = $24x + 2x^3$
CSO AG (be convinced)

A1 2

(c) $\frac{dy}{dx} = 24 + 6x^2$
A power of x decreased by 1

M1

For st. pt. $24 + 6x^2 = 0$

A1

Not possible since $24 + 6x^2 > 0$
Any valid explanation

E1 3

[10]

Q26.

(a) (i) p(3) = 27 – 45 + 21 – 3
Finding p(3)

M1

p(3) = 0 $\Rightarrow$ $x - 3$ is a factor

Shown = 0 plus a statement

Or $(x - 3)(x^2 - 2x + 1)$

A1 **2**

(ii) B is point (3,0)

Must have coordinates

B1 **1**

(b) (i) $\frac{dy}{dx} = 3x^2 - 10x + 7$

One term correct

M1

All correct with NO + c etc

A1 **2**

(ii) $3x^2 - 10x + 7 = 0$

Putting their $\frac{dy}{dx} = 0$

M1

$\Rightarrow (x - 1)(3x - 7) = 0$

Attempt to use quad formula or factorise

m1

$\Rightarrow$ at M, $x = \frac{7}{3}$

***CSO** factors correct etc*

A1 **3**

(c) $\frac{d^2y}{dx^2} = 6x - 10$

ft their $\frac{dy}{dx}$

B1ft

sub $x = 1$, $\Rightarrow \frac{d^2y}{dx^2} = -4$

ft their $\frac{d^2y}{dx^2}$

B1ft

2

(d) (i) $\frac{x^4}{4} - \frac{5x^3}{3} + \frac{7x^2}{2} - 3x \quad (+c)$

Increase one power by 1

M1

One term correct

A1

Two other terms correct

A1

All correct (condone missing + c)

A1

4

(ii) Realisation that limits are 0 and 1

Condone wrong way round

B1

$\left[\frac{1}{4} - \frac{5}{3} + \frac{7}{2} - 3\right] - 0$

Attempt to sub their limits into their (d)(i)

M1

$= -\frac{11}{12}$

CSO. *Must use F(1) – F(0) correctly*

A1

Area = $\frac{11}{12}$

CSO. *Convincing argument*

E1

4

[18]

Q27.

(a) (i) $\sqrt{x} = x^{\frac{1}{2}}$

Accept p = 0.5

B1

1

(ii) $\int \sqrt{x}\, dx = \frac{x^{1.5}}{1.5} \{+c\}$

Index raised by 1

M1

Correct ft on p. Condone missing '+c'

A1ft

2

(iii) Area = $\int_1^4 \sqrt{x}\, dx$

Limits 1 and 4 PI

B1

........= $\frac{4^{1.5}}{1.5} - \frac{1}{1.5}$

F(4) – F(1)

M1

$= \frac{14}{3}$

Accept 4.66 or better

A1

3

(b) (i) $y = x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$

Index (p – 1) ft

M1

When x = 4, y' (4) = 0.25

Attempt to find y'(4)

M1

When $x = 4$, $y = 2$

B1

Equation of tangent: $y - 2 = \frac{1}{4}(x - 4)$

accept other forms

A1

4

(ii) When $x = 0$, $y = 1$ B (0, 1)

*Subs x= 0 **and** then y = 0 into equation of tangent. PI*

M1

When $y = 0$, $x = -4$ $A(-4, 0)$

Correct ft y_B and x_A
(may be awarded as part of area calculation)

A1ft

Area = 0.5(1)(4) = 2

ft wrong sloping tangent and max of 1 further slip. Final answer must be +'ve

A1ft

3

(c) Translation

'Translation'/'translate(d)'

B1

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Accept equivalent in words provided linked to 'translation/move/shift'
(Note:B0B1 is possible)

B1

2

(d) $h = 1$

PI

B1

Integral = $h/2$ {…….}
{….} = f(1) + 2[f(2) + f(3)] + f(4)

OE summing of areas of the three traps

M1

$\{\ldots\} = 0 + 2(1 + \sqrt{2}\ ) + \sqrt{3}$

Condone 1 numerical slip

A1

Integral = $\frac{1}{2}$ {2 (1 + 1.414…) + 1.732}

Integral = 0.5 × 6.560… = 3.28 to 3sf

Must be 3.28

A1CAO 4

[19]

Q28.

(a) .... $= x^5 - x^{-3}$

One power correct

M1

Accept $p = 5$, $q = -3$

A1 2

(b) (i) $f'(x) = 5x^4$

ft on px^{p-1}

B1ft

$+3x^{-4}$

ft on $-qx^{q-1}$ provided $q < 0$

B1ft 2

(ii) $f'(x)\left\{= 5x^4 + \frac{3}{x^4}\right\} > 0$

M1 Considers sign of f'(x); a statement "f'(x) > 0 OE" with 'f increasing'.

M1

$\Rightarrow$ f is increasing {function}

A1 needs f'(x) of the form $ax^4 + \frac{b}{x^4}$, where a and b both > 0 and no incorrect *statements based on f'(x) at different values of x*

A1

2

(c) At (1, 0), f'(1) = 5 + 3 = 8

Attempts to find f'(1)

M1

$\Rightarrow$ grad. of normal = $-\frac{1}{8}$

Use of $m \times m' = -1$ PI

m1

ft on wrong f'(x)

A1ft

3

[9]

Q29.

(a) $\frac{1}{4}e^{4x}$

B1

1

(b) $\int e^{4x}(2x + 1)\,dx$

$u = 2x + 1$ $\quad$ $dv = e^{4x}$

by parts

M1

$du = 2$ $\quad$ $v = \frac{1}{4}e^{4x}$

$= \frac{1}{4}(2x + 1)e^{4x} - \frac{1}{2}\int e^{4x}\,dx$

Their ($uv - \int v du$)

M1

$= \frac{1}{4}(2x + 1)e^{4x} - \frac{1}{8}e^{4x}\ (+c)$

A1

3

(c) $u = 1 + \ln x$

$\frac{du}{dx} = \frac{1}{x}$ or $\frac{dx}{du} = e^{u-1}$

B1

$\int = \int u \, du = \frac{u^2}{2}$ (+c)

in terms of u only

M1A1

$= \frac{(1+\ln x)^2}{2} (+c)$

A1

4

[8]

Q30.

(a) (i) $a = \frac{dv}{dt}$

$= 12t + 8e^{-4t}$ m s^{-2}

M1 for either term correct

M1A1

2

(ii) When $t = 0.5$, $a = 6 + 8 \times e^{-2}$

m1

$= 7.08$ m s^{-2}

Condone 7.07

SC1 for 7.1 with no working

A1

2

(b) Using $F = ma$:

$F = 4 \times 7.08$

$= 28.3$ N

Ft from value awarded A1

B1ft

1

(c) $r = \int v dt$

At least two terms correct

M1

$= 2t^3 + \frac{1}{2}e^{-4t} + 8t + c$

Does not need + c

A1

When $t = 0$, $r = 0 \rightarrow c = -\frac{1}{2}$

Does not need $c = -\frac{1}{2}$

m1

$r = 2t^3 + \frac{1}{2}e^{-4t} + 8t - \frac{1}{2}$

Need r, s (or words)

A1

4

[9]

Q31.

(a) $\mathbf{a} = \frac{\mathrm{d}v}{\mathrm{d}t}$

$\mathbf{a} = -8e^{-2t}\,\mathbf{i} + (6 - 6t)\mathbf{j}$

*M1: Differentiating with either of the two components correct. Do not need to see **i** or **j**.*

*A1: Correct **i** component.*

*A1: Correct **j** component.*

M1
A1
A1

3

(b) (i) Using $\mathbf{F} = m\mathbf{a}$

$\mathbf{F} = 5 \times \{-8e^{-2t}\,\mathbf{i} + (6 - 6t)\mathbf{j}\}$

$= -40e^{-2t}\,\mathbf{i} + (30 - 30t)\mathbf{j}$

M1: Multiplying their acceleration by 5, even if not a vector.

A1: Correct expression.

M1
A1

2

(ii) Magnitude of $\mathbf{F}$ is

$\{(-40)^2 + (30)^2\}^{\frac{1}{2}}$

M1: Finding magnitude from two non-zero terms. Must add terms and square root. Condone $\{(40)^2 + (30)^2\}^{\frac{1}{2}}$

M1

= 50

A1: Correct answer only.

In this part, condone lack of negative signs in expression for force in (b) (i).

A1

2

(c) When $\mathbf{F}$ acts due west, $\mathbf{j}$ component is zero

$30 - 30t = 0$

M1: Putting $\mathbf{j}$ *component equal to zero.*

M1

$t = 1$

A1: Correct time.

A1

2

(d) $\mathbf{r} = -2e^{-2t}\,\mathbf{i} + (3t^2 - t^3)\mathbf{j} + \mathbf{c}$

M1: Integration with either of the two components correct. Do not need to see $\boldsymbol{i}$ *or* $\boldsymbol{j}$*.*

A1: Correct $\boldsymbol{i}$ *component.*

A1: Correct $\boldsymbol{j}$ *component.*

Condone lack of + $\boldsymbol{c}$

M1
A1
A1

When $t = 0$, $\mathbf{r} = 6\mathbf{i} + 5\mathbf{j}$ $\therefore \mathbf{c} = 8\mathbf{i} + 5\mathbf{j}$

dM1: Finding $\boldsymbol{c}$ *using 6*$\boldsymbol{i}$ *+ 5*$\boldsymbol{j}$ *and* $e^0 = 1$*.*

dM1

$\therefore \mathbf{r} = (8 - 2e^{-2t})\mathbf{i} + (5 + \frac{1}{4}t^4 - t^3)\mathbf{j}$

A1: Correct position vector.

A1

5

[14]

Q32.

$v = \dfrac{dt}{dv}$

M1 for either $\dfrac{ds}{dt}$ *or 1 of 2 terms correct*

(ignore signs)

M1

$= 10t - 12 \sin 4t$

A1A1

[3]

Q33.

(a) $\mathbf{a} = \dfrac{d\mathbf{v}}{dt} = (3t^2 - 15)\mathbf{i} + (6 - 2t)\mathbf{j}$

*A1 (**i** terms) A1 (**j** terms)*

M1A1A1

3

(b) (i) Using $\mathbf{F} = m\mathbf{a}$:
Force $= 4 \times \{(3t^2 - 15)\mathbf{i} + (6 - 2t)\mathbf{j}\}$

M1

$= (12t^2 - 60)\mathbf{i} + (24 - 8t)\mathbf{j}$

AG

A1

2

(ii) When $t = 2$, force $= -12\mathbf{i} + 8\mathbf{j}$

M1A1

Magnitude of force $= \sqrt{12^2 + 8^2}$ N

M1

$= 14.4$ (N)

A1

4

[9]

Q34.

(a) (i) $a = \dfrac{dv}{dt} = 6t - 6\cos 3t$

M1 for at least one term correct

M1A1

2

(ii) When $t = \frac{\pi}{3}, a = 6 \times \frac{\pi}{3} - 6\cos(3.\frac{\pi}{3})$

M1

$= 2\pi + 6$

AG

A1

2

(b) $r = t^3 + \frac{2}{3}\cos 3t + 6t + c$

M1 for 3 terms including cos3t term
Condone no '+ c'

M1A1

When $t = 0$, $r = 0$ $\therefore c = -\frac{2}{3}$

M1

$\therefore r = t^3 + \frac{2}{3}\cos 3t + 6t - \frac{2}{3}$

A1

4

[8]

Q35.

(a) $\mathbf{v} = \frac{\mathrm{d}r}{\mathrm{d}t}$

$\mathbf{v} = (3t^2 - 6t)\mathbf{i} + (4 + 2t)\mathbf{j}$

M1A1

2

(b) (i) $\mathbf{a} = (6t - 6)\mathbf{i} + 2\mathbf{j}$

M1A1ft

Using $\mathbf{F} = m\mathbf{a}$:
$\mathbf{F} = (18t - 18)\mathbf{i} + 6\mathbf{j}$

A1ft

3

(ii) When $t = 3$, $\mathbf{F} = 36\mathbf{i} + 6\mathbf{j}$

Magnitude is $\sqrt{36^2 + 6^2}$

M1

$= 36.5$

Accept $6\sqrt{37}$ *; ft from (b)(i)*

A1ft

2

(c) When **F** acts due north:
Component of **F** in the **i** direction is 0

M1

$18t - 18 = 0$
$t = 1$

ft from (b)(i)

A1ft

2

[9]